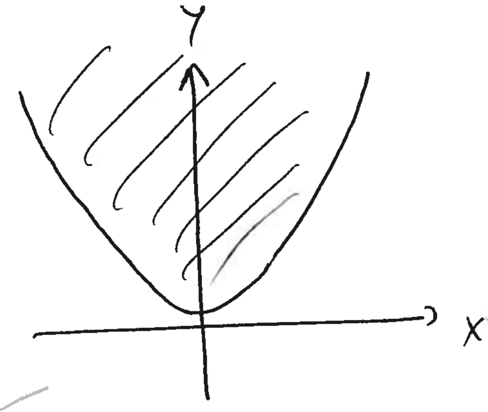
§ 12.1 Functions of two or more variables

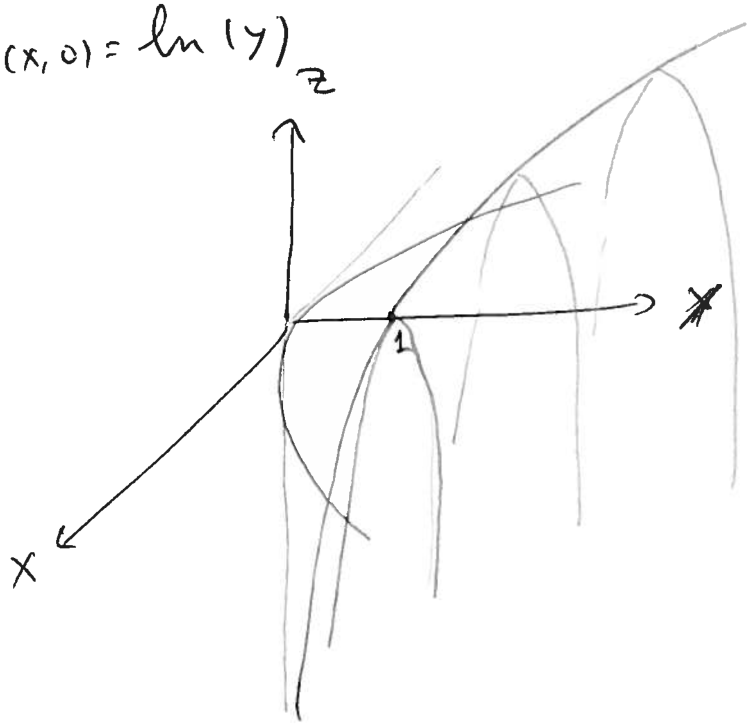
Ex $f(x,y) = \ln(y - x^2)$



Domain: $y - x^2 > 0 \qquad y > x^2$

"$\ln 0 = -\infty$"

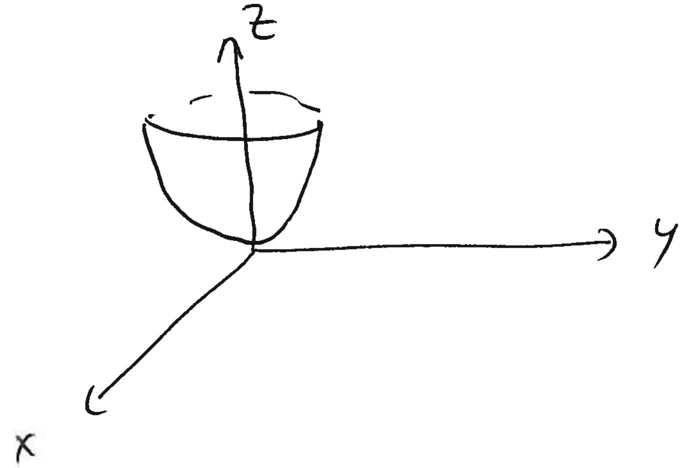
$x = 0 \Rightarrow f(x,0) = \ln(y)$



Hard to draw

Ex $f(x,y) = x^2 + y^2$ easier. Domain $\mathbb{R}^2$

$z = x^2 + y^2$
paraboloyd

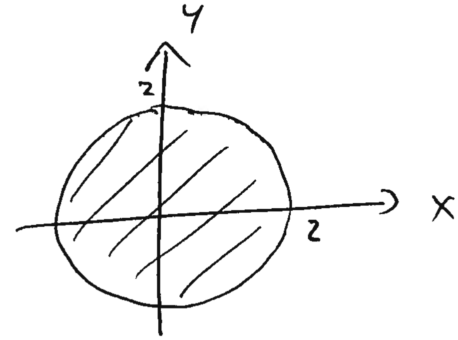


Ex $\quad f(x,y) = 2\sqrt{4-x^2-y^2}$

Domain $\subset \mathbb{R}^2$

$4 - x^2 - y^2 \geq 0$

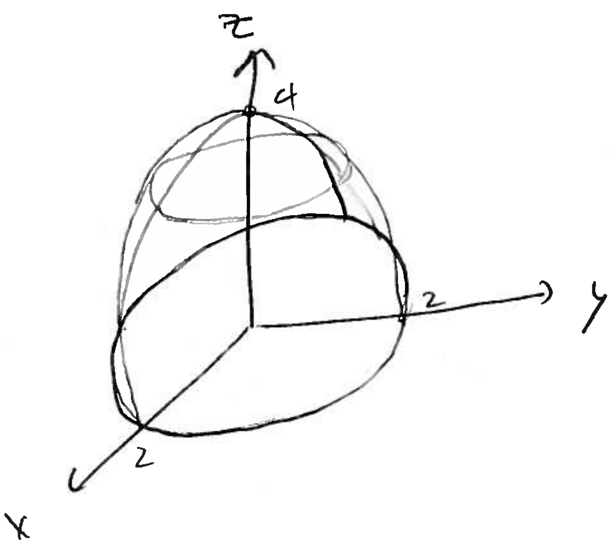
$4 \geq x^2 + y^2$



$z = 2\sqrt{4-x^2-y^2}$

$\frac{z^2}{4} = 4 - x^2 - y^2$

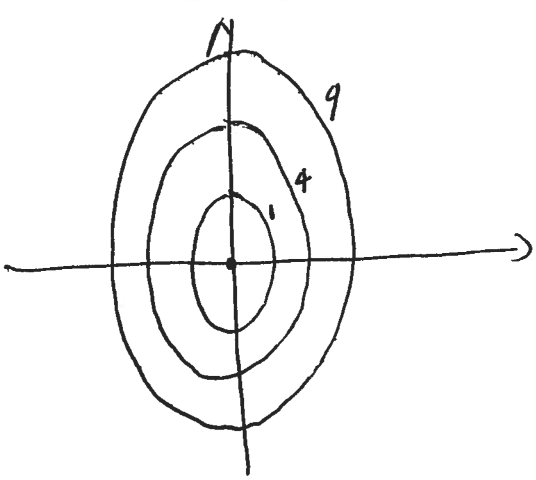
$4 = x^2 + y^2 + \frac{z^2}{4}$ top half of ~~sphere~~ ellipsoid



Level curves of $z = f(x,y)$

set $z = c$ so intersect with plane at height c

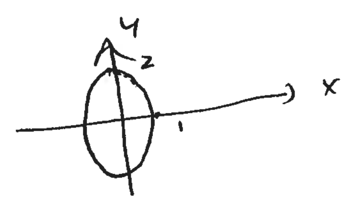
Eg $\quad f(x,y) = x^2 + \frac{y^2}{4}$



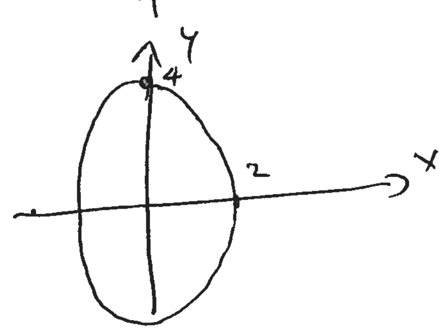
$c < 0 \qquad \emptyset$

$c = 0$

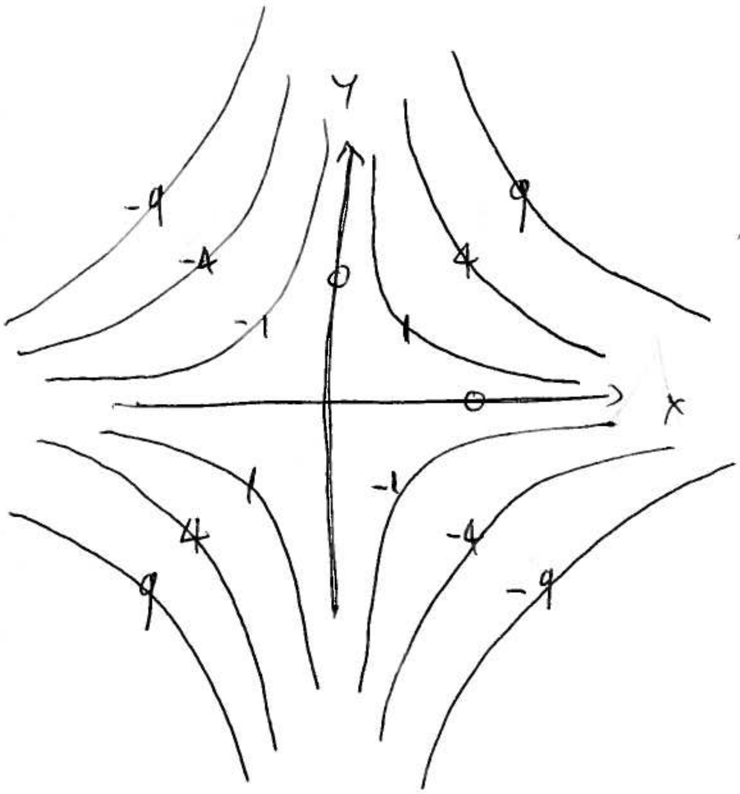
$c = 1$



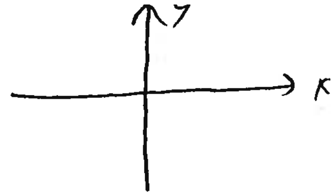
$c = 4$



Ex $f(x,y) = xy$

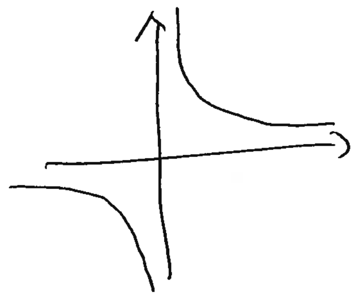


$c=0$



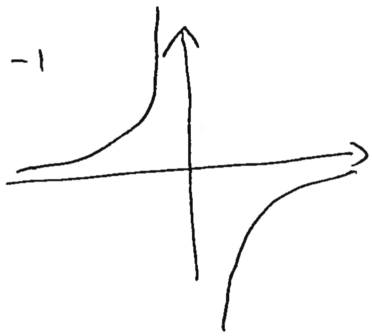
x & y axis

$c=1$



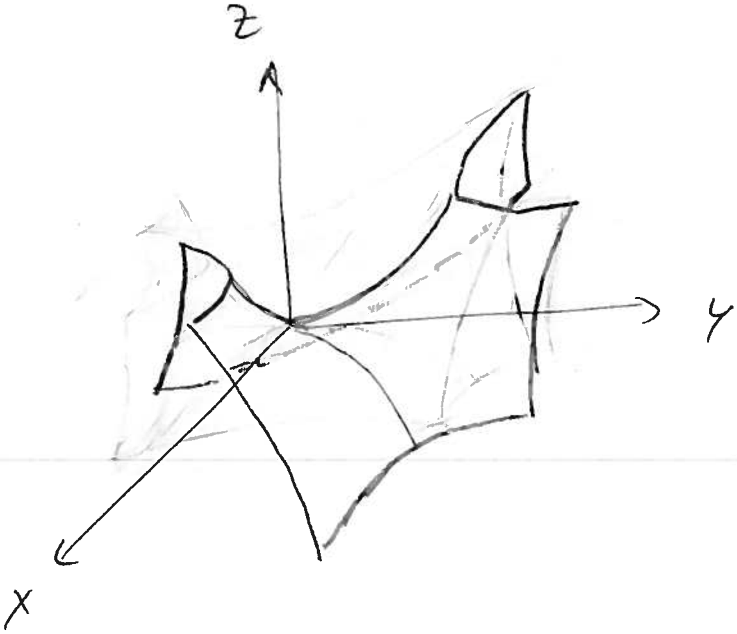
$xy = 1$

$c=-1$



$xy = -1$

$z = xy$



Saddle

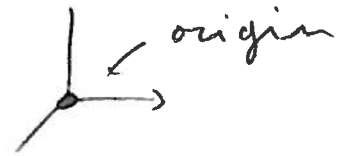
$x=y \quad z=x^2$

$x=-y \quad z=-y^2$

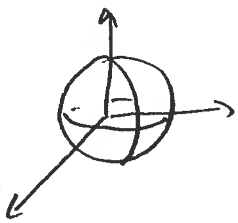
Ex $f(x,y,z) = x^2+y^2+z^2$

$c<0$ ϕ

$c=0$

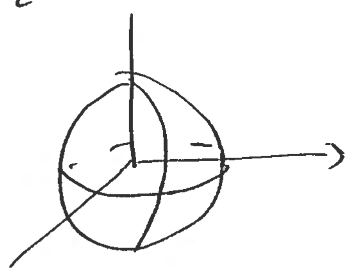


$c=1$



sphere of radius 1

$c=2$



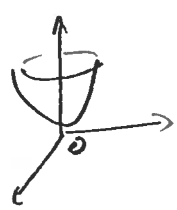
spere of radius $\sqrt{2}$

Can't graph $\cancel{w} = x^2+y^2+z^2$

Level surfaces are as good as it gets

Eg $f(x,y,z) = z - x^2 - y^2$

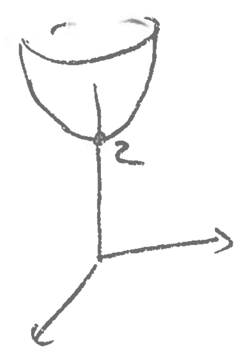
$c=0$



$c=1$

$z = x^2+y^2+1$

$c=2$



$c=-1$

$z = x^2+y^2-1$

$c=-2$